

NGUYEN LE GIA HUY

Software Engineer / Full-Stack & AI Systems

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PROFESSIONAL SUMMARY

Software Engineering graduate from HUTECH specialized in **Full-Stack Architecture, Distributed Systems & AI/RAG Integration**. Hands-on experience architecting enterprise monorepos, microservices, row-level concurrency locking, and offline-first mobile synchronization. Proficient in **Next.js, NestJS, Flutter/Isar, PostgreSQL (Neon pgvector), Redis, Docker, and Google Gemini API. Open to Internship and Full-Time Software Engineering opportunities.**

TECHNICAL SKILLS

Languages:	TypeScript, JavaScript, Dart, Python, SQL, C#, Java, HTML5/CSS3
Frontend & Mobile:	Next.js App Router, React, Flutter, Tailwind CSS, SCSS, Zustand
Backend & Database:	NestJS, Express.js, PostgreSQL (pgvector), Prisma ORM, Redis, Socket.IO
AI & DevOps:	Gemini API (RAG), TensorFlow.js (WebGL), Docker, Nginx, Playwright E2E, CI/CD

FEATURED PROJECTS

03/2026 – Present

HyperGlot — AI-Driven Japanese Language Learning Ecosystem (Monorepo)

Role: **Lead Full-Stack / Mobile Architect** | Monorepo: **github.com/nlgiahuy01112003/HyperGlot**

`Turborepo` `Next.js` `NestJS` `Flutter` `Isar DB` `Neon pgvector` `TensorFlow.js` `Playwright` `Docker`

Enterprise-grade Japanese language ecosystem for JLPT N5-N1 learners: learner web portal, content admin, offline mobile client, and Chrome extension under a strict Zero-Cost API Strategy.

- **Turborepo Monorepo Architecture:** Integrated `apps/web` (Next.js), `apps/admin` (TanStack Table & Recharts), `apps/mobile` (Flutter/Riverpod), `apps/api` (NestJS), and `packages/database` (Prisma) with shared type contracts.
- **Edge AI & Zero-Cost NLP:** Implemented kanji stroke recognition using TensorFlow.js with WebGL hardware acceleration and client-side morphological tokenization via kuromoji.js — eliminating cloud latency and third-party API costs entirely.
- **Semantic Vector Search:** Replaced costly managed vector databases with native PostgreSQL `pgvector` extension on Neon Serverless for semantic question indexing and retrieval.
- **Offline-First Mobile SRS:** Built local Isar DB caching on Flutter, allowing students to study spaced-repetition (SRS) flashcards smoothly without internet; synced state asynchronously upon reconnect.
- **Quality Assurance & Security:** Authored 60+ automated test cases, Playwright E2E cross-browser suite, achieved WCAG 2.1 AA accessibility compliance, and passed OWASP Top 10 security audits.

09/2025 – 03/2026

TechStore (Vanfio) — Concurrency-Controlled Electronics E-Commerce

Role: **Full-Stack Developer** | Demo: **website-for-aunt.vercel.app** | Source: **github.com/nlgiahuy01112003/techstore**

`Next.js` `React` `NestJS` `PostgreSQL` `Prisma ORM` `Redis` `Socket.IO` `MoMo` `Docker Compose`

Production electronics commerce platform featuring Next.js storefront, NestJS REST API, Socket.IO live support gateway, and strict inventory protection.

- **Concurrency Control & Row-Level Locking:** Enforced PostgreSQL `SELECT FOR UPDATE` locks inside database transactions, guaranteeing zero overselling when multiple customers checkout the last item concurrently.
- **PostgreSQL-Native Trigram Search (pg_trgm):** Implemented full-text ranking and trigram similarity with VI/EN/ZH synonym expansion, eliminating Meilisearch RAM footprint with sub-50ms search latency.
- **Idempotent Payment Webhooks:** Verified MoMo/VNPay webhooks with HMAC-SHA256 signatures backed by idempotency keys, preventing duplicate financial transactions under network retries.
- **Performance Optimization & Caching:** Deployed Redis caching for session tokens and high-traffic catalogs, reducing API latency by 45%; achieved 95+ Google Lighthouse score (CLS = 0).
- **Multi-Layer Defense-in-Depth:** Guarded by rotating JWT sessions, TOTP MFA for staff, RBAC access guards, strict DTO ValidationPipe, and Helmet brute-force/DDoS rate limiting.

OTHER CORE PROJECTS

08/2026 – 09/2026

AI Mock Exam Platform — SFIA Level 3-4 Assessment Platform

Role: **Full-Stack / AI Engineer** | Demo: **ai-exam-live** | Source: **GitHub**

Next.js | NestJS | Pyodide WebAssembly | Gemini AI Grader | Neon PostgreSQL (SSL) | MathJax | Web Audio API

- Built a full-stack mock exam & coding assessment platform for AI Engineers & Data Scientists aligned with SFIA Level 3-4 standards featuring a **1,020+ question bank** (Probability, ML, Transformers, MLOps).
- Integrated **Pyodide WebAssembly** for zero-cost in-browser Python/NumPy sandbox execution, **MathJax** for LaTeX equations, and synthesized audio cues via **Native Web Audio API**.
- NestJS backend connected to Neon PostgreSQL Serverless with automated **Offline Fallback** and **Gemini AI Code Grader & Smart Mentor** for automated code evaluation.
- Authored a comprehensive test suite of **24+ automated API test cases** covering grading algorithms, randomized question shuffling, and AI Smart Mentor insights (100% Passed).

06/2026 – 07/2026

OptiBot RAG System (KB-Harvest) — Grounded AI Support Assistant & ETL

Role: **Python / AI Engineer** | Source: **github.com/nlgiahuy01112003/kb-harvest**

Python | Google Gemini API | BeautifulSoup4 | SHA-256 Delta Sync | Docker | GitHub Actions Cron

- Automated RAG pipeline: scraped Zendesk Help Center via paginated REST API, engineered a multi-pass HTML-to-Markdown cleaner (saving 40%+ LLM context tokens).
- Devised a **SHA-256 delta sync** algorithm that calculates content hashes, uploading only new or modified articles and eliminating **~85% of redundant daily uploads**.
- Integrated **Gemini File Search** (512-token chunks with 100-token overlap) for terminal Q&A support answers grounded in official documentation with direct URL citations.
- Dockerized single-command execution automated via daily **GitHub Actions cron (02:00 UTC)** archiving run-logs, sync state, and markdown artifacts.

03/2024 – Present

POMI Milk Tea — Real-Time Multi-Platform F&B Ordering & AI Inventory System

Role: **Backend Developer** | Source: **github.com/nlgiahuy01112003/pomi_web**

React | Express.js | PostgreSQL | Socket.IO | Flutter | Gemini AI & Zalo Alert | Goong.io

- Engineered Express TypeScript REST API with bidirectional Socket.IO, synchronizing order states to kitchen displays within 800ms with Idempotency Key **order deduplication**.
- Built a **Recipe & Ingredient Inventory Engine** (separating Base drinks vs. Toppings with automated per-cup deduction), integrated a **Gemini AI Assistant** for admins, and automated **Zalo low-stock alerts**.
- Integrated Goong.io Map APIs to identify nearest store branches and automatically calculate delivery distances and dynamic fees.

EDUCATION

09/2021 – 03/2026

Bachelor of Engineering in Information Technology (Software Engineering)

Ho Chi Minh City University of Technology (HUTECH) — GPA: 2.86/4.0

- **Core Foundations:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Relational Databases, Operating Systems, Computer Networking, Software Architecture, and Automated Testing.
- **Practical Projects:** Led agile software project teams for web and mobile applications, designing database schemas, API contracts, Git version workflows, and CI/CD automation.

CERTIFICATIONS & LANGUAGES

- **English CEFR B1 — HUTECH:** General English Test at Intermediate Level; issued Dec 2025.
- **Udemy — Flutter Development Bootcamp with Dart:** Dr. Angela Yu; Dart OOP, Flutter State & APIs.
- **Communication & Teamwork Skills — HUTECH:** Issued Apr 2025; technical problem solving and team alignment.
- **Udemy — The Complete Full-Stack Web Development Bootcamp:** Dr. Angela Yu; React, Node.js, REST API.
- **Udemy — 100 Days of Code™ Python Pro Bootcamp:** Dr. Angela Yu; Python OOP, Web Scraping, Automation.

WORK EXPERIENCE

12/2025 – 06/2026

IT Helpdesk / Manual QA Tester & System Deployment

EMPOWER SC

- Operated and supported the **Dimchan All-in-One** POS and retail ecosystem; categorized software, printer, and network issues with a > 95% resolution success rate.
- Conducted manual E2E testing across Web/App interfaces; documented reproducible bug tickets and collaborated with development teams to resolve root causes.

08/2025 – 10/2025

IT Support & System Deployment Specialist (Project Contract)

HCMC Digital Transformation Center (DXCENTER)

- **Business Process Training (First 2 Weeks):** Recruited for web technical expertise under a municipal digital project; mastered end-to-end workflows of 2 public portals (motcua.tphcm.gov.vn, qlvb.tphcm.gov.vn).
- **Field Stationed at Binh Khanh Ward (1.5 Months):** Directly trained civil servants and citizens on both portals; handled user inquiries, reproduced edge-case bugs, and dispatched structured bug reports to L2 Developers.
- **Hardware, Networking & Digital Signatures:** Troubleshot PC hardware, office LAN networks, and configured digital authentication/signatures; fulfilled deployment contract milestones.